

Yash Chitambar

yash_chitambar@berkeley.edu | (408) 781-8163 | [linkedin.com/yash-chitambar](https://www.linkedin.com/yash-chitambar)

EDUCATION

University of California, Berkeley

B.A. Computer Science & Data Science, Domain Emphasis in Robotics

Class of 2028, GPA: 3.4

Coursework: Deep Reinforcement Learning, Machine Learning, Artificial Intelligence, Efficient Algorithms & Intractable Problems, Computer Security, Probability & Random Processes, Discrete Mathematics & Probability Theory, Data Structures, Principles & Techniques of Data Science, Circuits & Devices, Linear Algebra, Multivariable Calculus, Physics

Skills: Python, Java, C/C++, Go, MATLAB, PostgreSQL, GCP/AWS, Docker/Kubernetes, Trino/Spark, Git

ML & Robotics Frameworks: PyTorch, JAX, NumPy/Pandas, CUDA, ROS2, Isaac Sim, NVIDIA Newton, MuJoCo, Linux

PROFESSIONAL EXPERIENCE

Salesforce

San Francisco, CA

Machine Learning Engineering Intern - AI Infrastructure (CECM Team)

May 2026 — August 2026

- Architected **semantic data layer** on **petabyte-scale** data-lake for LLM agents. Designed Pydantic annotation model & build-time Cube config generator into **NLP SQL compiler MCP** for deterministic agent SQL generation
- Integrated semantic layer into **production data DAGs** & A/B tested against agent generated SQL MCP baseline, achieving **7x faster query** response, **96% reduction** in token usage, & **3.3x reduction** in **Trino API calls**
- Created **evaluation pipeline** for migration of **forecasting foundation models** (Moirai/Prophet) from EC2 Airflow DAGs to **AWS SageMaker**, benchmarking throughput and cost metrics across **GPU accelerated instances**

Selector.ai

Santa Clara, CA

Machine Learning Engineering Intern

November 2025 — May 2026

- Built real-time ML pipelines for providing **real-time textual summarization** of timeseries data dashboards (metrics, logs, traces, & maps) scaling to **10M+ data points/day** and supporting **150+ active users** in production.
- Trained models including **TS Random Forrest**, **Transformer(<1B)**, **GNNs**, & **UMAP/DBSCAN** for metric insight generation, achieving **94–96% accuracy** on benchmark time series analysis for summarization accuracy
- Developed end-to-end serving pipelines in **PyTorch**, containerized with **Docker** & orchestrated with **Kubernetes APIs**, achieving **<120 ms latency** and handling **5K+ concurrent requests**

Autodesk | Berkeley AI Research Lab (Autolab)

Berkeley, CA

Undergraduate Machine Learning Researcher - accepted to ICRA 2026/Science Robotics May 2025 — December 2025

- Contributed to large-scale **benchmarking cloud infra** ([M-NET](#)) & **eval standardization** project for real-world robot manipulation tasks (cable routing, grasping, placement) & **embodied reasoning**
- Worked on **decentralized server-client evaluation** architecture that cryptographically verifies submitted video/execution logs against live eval protocols for global authenticated leader board for robot manipulation tasks
- Developed **novel robot cable routing** algorithm ([CRAFT](#)) improving on prior **SOTA by 54.2%** route accuracy

Nvidia & Intuitive Surgical | Berkeley AI Research Lab (Autolab)

Berkeley, CA

Undergraduate Machine Learning Researcher

July 2026 — Present

- Architecting **agentic harness** for surgical robotics that automatically discovers, composes, & chains manipulation skills (grasp, reach, cut) from unsegmented demonstrations on **simulated DVRK** in **NVIDIA Newton/Isaac Sim**
- Fine-tuned **GR00T N1.5 VLA** policy via **LoRA** on **300 DVRK PSM demonstrators**, outputting 16 DOF joint trajectories from single-camera & proprioceptive input, reaching target within **1mm accuracy** across eval demos
- Developing **RL fine tuning pipeline** (GAP, EMPIRE) to refine agentic skill discovery primitives beyond imitation based learning baselines in **NVIDIA Newton physics engine**

PROJECTS

Dexterous Hand Manipulation Policy (PyTorch, Mujoco/MFX, JAX, DINOv2, PPO)

- Trained PPO, diffusion, & BC + RL policies for dexterous hand cube reorientation in MuJoCo, comparing pure RL imitation on **800 expert** demonstration, achieving **91% policy success** on **asymmetric actor-critic PPO** policy
- Implemented **diffusion policy** from scratch through **film-conditioned 1D U-net** with EMA, cosine noise scheduling, & receding-horizon control with **80.9% success** with DDIM accelerated inference
- Developed two-camera visuomotor pipeline from **frozen DINOv2 features**, spatial-softmax keypoints, behavior cloning, & RECAP-Inspired PPO refinement with KL regularization showing regulation reduced policy drift

Dreamer 4-Style World Model for Robot Hand Manipulation (PyTorch, MuJoCo, Flow Matching, Transformers)

- Implemented Dreamer-style **world model** from scratch via causal masked **video tokenizer** plus a block-causal **dynamics transformer** (GQA, RoPE, QK-Norm) trained with flow matching at **13M & 40M parameters**
- Built **MuJuCo SO-101 simulator** with IK-scripted experts for 4 tasks and collected **311K frames**. Model predicted next 8 frames with **11x lower error** than baseline, used 4-step sampling to keep **94% model accuracy overall**
- Validated world model controllability, representation and rollout fidelity. **Reversing input actions** shifts prediction by 6x for models error, frozen latents decode gripper position & imagined rollouts match token reconstruction

ADDITIONAL

Leadership: VP of Projects, DataStory Consulting — *lead student org in delivering data-driven solutions for client projects*

Volunteering: Mentor FRC Robotics Team 3482 — *teach students CAD and advanced odometry techniques for robotics*

Hobbies: Triathlon, Running, CAD/3D Printing, Design, Jazz/Classical Piano, Mountain Biking, Soccer, & Football